



**Faculty of Science and Technology**

**Jagran School of Engineering**

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**Course Name: Web Application Development**

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**Submitted By:**

Name: Harshita Soni

JLUID: JLU09173

ROLL NO.: 2024BTCSE010

**Submitted To:**

Ms Reshma Chavan

# **1. Project Title: Ability Jobs – An Inclusive Job Portal for Persons with Disabilities**

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## **2. Abstract**

AbilityJobs is a web-based application developed to address the employment challenges faced by Persons with Disabilities (PWDs). The platform provides an accessible and inclusive environment where job seekers, employers, and NGOs can interact efficiently.

The system implements accessibility standards (WCAG 2.1 AA), smart job filtering, and an intuitive user interface to ensure usability for all users. Unlike traditional job portals, AbilityJobs is specifically designed for inclusivity, bridging the gap between opportunity and accessibility.

The project demonstrates how technology can be leveraged to promote social equality and improve employment outcomes for marginalized communities.

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## **3. Introduction**

Employment is a fundamental right, yet many PWDs face barriers due to lack of accessible systems. Traditional job portals do not cater to accessibility needs such as screen readers, keyboard navigation, or disability-specific job matching.

AbilityJobs aims to eliminate these barriers by providing a PWD-centric digital platform that ensures accessibility, usability, and inclusivity.

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## **4. Problem Statement**

- Low employment rate among PWDs
  - Lack of dedicated job platforms
  - Poor accessibility in existing systems
  - Limited employer awareness
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## **5. Objectives**

## Primary Objectives

- Develop an inclusive job portal for PWDs
- Ensure accessibility compliance (WCAG 2.1 AA)
- Provide job matching based on disability

## Secondary Objectives

- Enable employer participation
- Integrate NGO support systems
- Improve user engagement and tracking

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## 6. Scope of the System

The system includes:

- Job search and application tracking
- Employer job posting and management
- NGO support services

## Exclusions:

- No backend/database server
- No real-time communication

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## 7. Stakeholder Analysis

| Stakeholder | Role             | Benefits              |  |
|-------------|------------------|-----------------------|--|
| Job Seekers | Find jobs        | Equal opportunities   |  |
| Employers   | Hire talent      | Diversity & inclusion |  |
| NGOs        | Support users    | Social impact         |  |
| Society     | Inclusive growth | Reduced inequality    |  |

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## 8. System Architecture

### Architecture Type

Client-side Web Application

### Layers

1. **Presentation Layer** – UI (HTML, CSS, JS)
2. **Application Layer** – Logic handling
3. **Data Layer** – LocalStorage + JSON

### Key Features

- No backend dependency
  - Offline capability
  - Responsive design
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## 9. System Design

### User Flow

Register → Profile → Search Jobs → Apply → Track Status

### Design Principles

- Accessibility-first
  - User-friendly interface
  - Minimal complexity
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## 10. Module Description

### 1. Authentication Module

- User registration
- Input validation
- Session handling

## **2. Profile Management**

- Resume upload
- Profile completion indicator
- QR code sharing

## **3. Job Portal**

- Job listings (500+)
- Disability filters
- Save jobs

## **4. Application Tracking**

- Status updates
- Interview alerts
- History logs

## **5. Employer Module**

- Job posting
- Candidate filtering
- Quota compliance

## **6. NGO Module**

- Training programs
- Counseling services
- Resource hub

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## **11. Database Design**

### **Storage Method**

- Browser LocalStorage

## Data Format

- JSON

## Entities

- Users
- Jobs
- Applications
- NGOs

## Sample Structure

```
User{  
  id,  
  name,  
  disability,  
  skills,  
  resume  
}
```

```
Job {  
  id,  
  title,  
  company,  
  requirements  
}
```

```
Application {  
  user_id,  
  job_id,  
  status  
}
```

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## 12. Technology Stack

### Frontend

- HTML5
- CSS3

- Tailwind CSS
- JavaScript (ES6+)

## Accessibility

- WCAG 2.1 AA
- ARIA labels
- Keyboard navigation

## Data Handling

- LocalStorage
- JSON

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## 13. Security Considerations

- Input validation
- Data privacy (client-side storage)
- Secure session handling
- Prevention of invalid inputs

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## 14. Testing Strategy

### Types of Testing

- Functional Testing
- UI Testing
- Accessibility Testing
- Cross-browser Testing

### Results

- High usability
- Smooth navigation
- Accessible interface compliance

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## 15. Deployment

- Browser-based deployment
- Deployed on Render
- Works offline and online on render service link
- Compatible with desktop & mobile

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## 16. Limitations and Future Enhancements

- No real-time updates
- AI-based job recommendation
- Video interview integration
- Mobile application
- Multi-language support
- Live chat system
- Credential verification

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## 17. Conclusion

AbilityJobs is an innovative solution that addresses employment challenges for Persons with Disabilities through accessible technology.

The system demonstrates how inclusive design and thoughtful implementation can create meaningful social impact and promote equal opportunities.

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## 18. References

- WCAG 2.1 Guidelines and geeks for geeks
- MDN Web Docs
- HTML, CSS, JavaScript Documentation on W3S school



